

2012 湖南省力学竞赛理论力学试题答案

一、概念题 (2 分*4=8 分)

1. $\frac{Fac}{\sqrt{a^2 + b^2 + c^2}}$

2. 7.07m/s

3. (1) $f_1 = \tan \varphi_1 \geq \frac{P}{Q_1}$; $f_2 = \tan \varphi_2 \geq \frac{F_2}{N_2} = \frac{P}{Q_1 + Q_2}$; (2) $\frac{d}{D} \leq \frac{P^2}{Q_1^2}$

二、(2 分*4=8 分) (1) $f_A \geq \tan \beta = \frac{AD}{ED} = \frac{3\sqrt{3}}{12 + \sqrt{3}}$;

(2) $f_A \geq \tan \beta = \frac{AD}{ED} = \frac{3\sqrt{3}}{12 + \sqrt{3} \tan 15^\circ} = \frac{3\sqrt{3}}{9 + 2\sqrt{3}}$, $f_3 \geq \frac{3\sqrt{3}}{9 + 2\sqrt{3}}$, $-b_{\min} \geq 2h \tan \beta = \frac{6\sqrt{3}}{9 + 2\sqrt{3}} h$

三、(2 分*4=8 分) (1) $x^2 y^2 - (a - y)^2 (a^2 - y^2) = 0$;

(2) $v_M = \sqrt{v_{Mx}^2 + v_{My}^2} = v_0 \sqrt{1 - 2 \sin^3 \varphi + \sin^4 \varphi}$ (以 v_0 和 φ 表示)

(3) $\omega = \frac{v_0 \sin^2 \varphi}{a}$; (4) $\alpha = \frac{v_0^2}{a^2} \sin^2 \varphi \sin 2\varphi$.

四、(3 分+4 分=7 分) $\omega_{AE} = \frac{v_{C'A}}{AC} = \frac{v \cos \alpha + u \sin \alpha}{l}$; $-\alpha_{AE} = \frac{2u(v + u \tan \alpha)}{l^2}$.

五、(2 分+3 分+2 分=7 分) (1) $2M - 3mga + \sqrt{2}Qa = 0$; (2) $\varepsilon_1 = \frac{3g}{4a}$; (3) $F_E = \frac{5\sqrt{2}}{8} mg$.

六、(2 分+3 分+2 分=7 分) (1) $v_C = \omega l \sin \theta$; (2) $\alpha = \frac{g \cos \theta (1 - 3 \sin^2 \theta)}{2l(1 + 3 \sin^2 \theta)}$; (3) $F_N = \frac{6mg(1 - 3 \sin^2 \theta)}{(1 + 3 \sin^2 \theta)}$.

七、(3 分+2 分+2 分=7 分) (1) $\alpha = \frac{3g}{8l}$; (2) $F_T = \frac{3\sqrt{3}}{16} P$; (3) $F_N = \frac{13}{16} P$.

八、(2 分*4=8 分) (1) $\omega = \dot{\varphi} = \sqrt{\frac{30g}{41R}} = 0.855 \sqrt{\frac{g}{R}}$; (2) $\omega = \dot{\varphi} = \sqrt{\frac{120g}{107R}} = 1.059 \sqrt{\frac{g}{R}}$;

(3) $\omega = \dot{\varphi} = \sqrt{\frac{150g}{157R}} = 0.9775 \sqrt{\frac{g}{R}}$; (4) $\omega = \dot{\varphi} = \sqrt{\frac{24g}{25R}} = 0.9798 \sqrt{\frac{g}{R}}$.

九、(2 分+3 分+3 分=8 分) (1) $\omega = \frac{3}{2} \frac{v}{l} \sin \theta$; (2) $\omega = \frac{6v \sin \theta}{l(1 + 3 \sin^2 \theta)}$; (3) $\omega = \frac{12v(\sin \theta - f \cos \theta)}{l(8 - 6 \cos^2 \theta - 3f \sin 2\theta)}$.

十、(3 分+4 分=7 分) (1) $v_C = \sqrt{\frac{3Rg \cos^2 \theta (1 - \cos \theta)}{2(1 + \sin^2 \theta)}}$; (2) $a_C = \frac{\sin 2\theta (5 + \sin^2 \theta) - 8 \sin \theta}{2(1 + \sin^2 \theta)^2} g$.

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一、1、 N_0 ; (2分) 2、 P (2分) 3、 $P = N_0 \left(1 + \frac{EA}{cl} \right)$ (3分)

二、(1) $\sigma_{t,\max} = \frac{4qL}{a}$; (2分) (2) $\Delta l_{AB} = \frac{2qL^2}{Ea}$; (2分) (3) $\tau = 0$ 。(3分)

三、(1); $\frac{2-\sqrt{3}}{2}h = 0.134h$ (2分) (2) $\frac{2-\sqrt{3}}{2} \frac{E_t b h^3}{M_z} = 0.134 \frac{E_t b h^3}{M_z}$; (3分)

(3) $(1+\sqrt{3}) \frac{M_z}{bh^2} = 2.732 \frac{M_z}{bh^2}$ 。(2分)

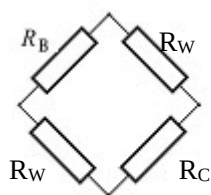
四、(1) $q_0 = \frac{2EI}{125l^3}$; (2分) (2) $\frac{EI}{500l}$; (1分) (3) $\frac{3EI}{1000l}$; (3分)

五、(1) $[M] = 0.659[\tau] \frac{\pi D^3}{16} = 0.1295[\tau] D^3$ 。(4分) (2) $\alpha \geq \frac{1}{2}$ (3分)

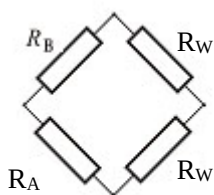
六、(1) $P = 6EIK$; (3分) (2) $y_B = \frac{EIKl^3}{4}$; (4分)

七、(1) $\tau_A = \frac{2}{3}\tau_s$; (2分) (2) $T_A/T = \frac{3}{53}$ 。(5分)

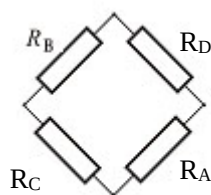
八、(1) 在如下三图中分别标出测量圆管中的内压 p 、 σ_M 和 τ 的应变片或补偿片布置图; (3分)



a.测量 p



b.测量 σ_M



c.测量 τ

(2) $p = \frac{4E\varepsilon_a t}{3(1-\mu)D}$; $\sigma_M = \frac{E\varepsilon_b}{2(1+\mu)}$; $\tau = \frac{E\varepsilon_c}{2(1-\mu)}$ 。(6分)

九、(8分, 每问各4分) (1) $\tan kl \tanh kl = 1$, $k^2 = \frac{P}{EI}$, $I = \frac{ab^3}{6}$

(2) $\tan kl \tanh kl = -1$, $k^2 = \frac{P}{EI}$, $I = \frac{ab^3}{6}$

十、(1) $p_0 = \frac{Eh}{R^2} \Delta$ (4分); (2) $p = \frac{Eh\Delta}{R^2(1-\mu)}$ (3分); (3) $N = \frac{2\pi Eh\Delta}{1-\mu}$ (3分)